

Viva Questions & Answers

Practicals 01–13 | Statistics & ML Foundations

Practical 01 – Matrix Operations (NumPy)

Q1. 1. What is the difference between element-wise multiplication and matrix multiplication?

Ans: Element-wise multiplication (using `*` or `np.multiply()`) multiplies corresponding elements of two matrices of the same shape, producing a matrix of the same shape. Matrix multiplication (using `np.dot()` or `@`) follows linear algebra rules where the number of columns of the first matrix must equal the number of rows of the second; it computes the dot product of rows and columns.

Q2. 2. Which NumPy function is used for matrix multiplication?

Ans: `np.dot(A, B)` or the `@` operator (`A @ B`) are used for matrix multiplication in NumPy. `np.matmul()` is also available and behaves similarly to `@`.

Q3. 3. What is the condition for addition of two matrices?

Ans: Two matrices can be added only if they have the same dimensions (same number of rows and columns). Each element of one matrix is added to the corresponding element of the other.

Q4. 4. What is the determinant of a matrix and what does it indicate?

Ans: The determinant is a scalar value computed from a square matrix. It indicates whether the matrix is invertible (non-zero determinant) or singular (zero determinant). It also represents the scaling factor of the linear transformation described by the matrix. Computed using `np.linalg.det()`.

Q5. 5. When does the inverse of a matrix exist?

Ans: The inverse of a matrix exists only when the matrix is square ($n \times n$) and its determinant is non-zero (i.e., the matrix is non-singular/invertible).

Q6. 6. Which NumPy function is used to find the inverse of a matrix?

Ans: `np.linalg.inv(A)` is used to compute the inverse of a square matrix `A`.

Q7. 7. What are eigenvalues and eigenvectors of a matrix?

Ans: For a square matrix `A`, a non-zero vector `v` is an eigenvector if $Av = \lambda v$, where λ (a scalar) is the corresponding eigenvalue. Eigenvectors represent special directions that only get scaled (not rotated) by the matrix transformation, and eigenvalues represent the scale factors.

Q8. 8. What is the significance of eigenvalues in data science or machine learning?

Ans: Eigenvalues are central to PCA (dimensionality reduction), indicating the amount of variance captured by each principal component. They are also used in spectral clustering, Google's PageRank algorithm, and stability analysis of machine learning models.

Q9. 9. What will happen if we try to find the inverse of a singular matrix?

Ans: If we try to invert a singular matrix ($\det = 0$), NumPy raises a `LinAlgError: Singular matrix` exception because a singular matrix has no inverse.

Q10. 10. What is the difference between `np.dot()` and `*` operator in NumPy matrices?

Ans: `np.dot()` performs true matrix multiplication (rows $\times$ columns dot product). The `*` operator performs element-wise multiplication, multiplying each element at the same position. For example, if `A` and `B` are 2×2 matrices, `A*B` gives a 2×2 result of element-wise products, whereas `np.dot(A,B)` gives the standard matrix product.

Practical 02 – Principal Component Analysis (PCA)

Q1. 1. What is the full form of PCA?

Ans: PCA stands for Principal Component Analysis.

Q2. 2. Why is data standardization important before applying PCA?

Ans: PCA is sensitive to the scale of features. If features have different units or ranges, those with larger magnitudes dominate the covariance matrix. Standardization (zero mean, unit variance) ensures all features contribute equally to the analysis.

Q3. 3. What is a covariance matrix?

Ans: A covariance matrix is a square symmetric matrix where each entry (i, j) represents the covariance between feature i and feature j . It captures how features vary together and is the basis for PCA.

Q4. 4. What do eigenvalues represent in PCA?

Ans: Each eigenvalue represents the amount of variance (information) captured by its corresponding principal component (eigenvector). Larger eigenvalues correspond to principal components that explain more variance in the data.

Q5. 5. What is meant by a principal component?

Ans: A principal component is a new variable formed as a linear combination of the original features. It points in the direction of maximum variance in the data. The first PC captures the most variance, the second PC (orthogonal to the first) captures the next most, and so on.

Q6. 6. Why are eigenvalues sorted in descending order in PCA?

Ans: Sorting eigenvalues in descending order ensures we select the principal components that explain the most variance first. This allows us to retain the most informative components and discard those with negligible variance.

Q7. 7. What is dimensionality reduction?

Ans: Dimensionality reduction is the process of reducing the number of features in a dataset while retaining as much information (variance) as possible. PCA achieves this by projecting data onto a lower-dimensional space defined by the top principal components.

Q8. 8. Which sklearn class is used to implement PCA in Python?

Ans: `sklearn.decomposition.PCA` is used. Example: `from sklearn.decomposition import PCA; pca = PCA(n_components=2); pca.fit_transform(X_scaled)`.

Q9. 9. What is explained variance ratio in PCA?

Ans: The explained variance ratio for each principal component is the proportion of the total dataset variance that the component captures. It is accessed via `pca.explained_variance_ratio_`. The sum across all selected components tells us how much total variance is retained.

Q10. 10. How do you verify that manual PCA results match sklearn output?

Ans: You compare the transformed data from both methods. Due to sign ambiguity (eigenvectors can be flipped), you compare absolute values or align signs before comparing. If values match (within floating-point tolerance), both implementations are correct.

Practical 03 – Naive Bayes Spam Classification

Q1. 1. What is Naive Bayes Classification?

Ans: Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem. It predicts the class of a new instance by computing the posterior probability for each class and choosing the class with the

highest probability.

Q2. 2. Why is Naive Bayes called 'naive'?

Ans: It is called 'naive' because it makes the strong (naive) assumption that all features (words) are conditionally independent of each other given the class label. In reality, words in text are not independent, but this simplification makes the algorithm computationally efficient and often works well in practice.

Q3. 3. What is Bayes' Theorem?

Ans: Bayes' Theorem states: $P(\text{Class}|\text{Features}) = [P(\text{Features}|\text{Class}) \times P(\text{Class})] / P(\text{Features})$. It provides a way to compute the probability of a class given observed features using prior probabilities and likelihoods.

Q4. 4. What are prior probabilities in spam classification?

Ans: Prior probability is the probability of a class before observing any features. In spam classification: $P(\text{Spam}) = (\text{number of spam emails}) / (\text{total emails})$ and $P(\text{Not Spam}) = (\text{number of non-spam emails}) / (\text{total emails})$.

Q5. 5. What are conditional probabilities?

Ans: Conditional probability $P(\text{word}|\text{class})$ is the probability of a word appearing given the class. For example, $P(\text{'free'}|\text{Spam})$ is the probability that the word 'free' appears in a spam email.

Q6. 6. What is Laplace smoothing and why is it used?

Ans: Laplace smoothing (add-1 smoothing) adds 1 to every word count to avoid zero probabilities. Without it, if a word in the test email never appeared in training data for a class, its probability would be zero, making the entire class probability zero regardless of other words.

Q7. 7. What is meant by vocabulary in text classification?

Ans: The vocabulary is the set of all unique words across all training emails. It defines the feature space used to represent emails as word-count vectors or frequency distributions.

Q8. 8. How do we calculate $P(\text{word}|\text{Spam})$?

Ans: $P(\text{word}|\text{Spam}) = (\text{count of 'word' in spam emails} + 1) / (\text{total words in spam emails} + \text{vocabulary size})$. The +1 and +|V| account for Laplace smoothing.

Q9. 9. What happens if a word is not present in the training dataset?

Ans: Without smoothing, $P(\text{word}|\text{class}) = 0$, making the product of all probabilities zero. With Laplace smoothing, unseen words get a small non-zero probability, preventing this zero-probability problem.

Q10. 10. Why is Naive Bayes suitable for spam email detection?

Ans: Naive Bayes is fast, works well with high-dimensional text data, requires relatively little training data, handles missing features gracefully, and despite its naive independence assumption, performs surprisingly well for text classification tasks like spam filtering.

Q11. 11. What is the role of tokenization in email classification?

Ans: Tokenization splits raw email text into individual words (tokens). This converts unstructured text into a countable set of features that the Naive Bayes model can process.

Q12. 12. How is the final class (Spam/Non-Spam) decided in Naive Bayes?

Ans: The class with the highest posterior probability wins: $\text{Class} = \text{argmax } P(\text{Class}) \times \prod P(\text{word}|\text{Class})$. In practice, log probabilities are used to avoid numerical underflow from multiplying many small probabilities.

Q13. 13. Which Python libraries can be used for implementing Naive Bayes classifiers?

Ans: scikit-learn provides GaussianNB, MultinomialNB, and BernoulliNB. NLTK supports text preprocessing. NumPy can be used for manual implementation. pandas is used for data handling.

Practical 04 – Coin Toss & Poisson Distribution

Q1. 1. What is experimental probability?

Ans: Experimental probability is the probability estimated from actual experiments or simulations. It equals (number of favorable outcomes observed) / (total number of trials). It varies with each experiment but converges to theoretical probability as the number of trials increases.

Q2. 2. What is theoretical probability?

Ans: Theoretical probability is the expected probability calculated using mathematical reasoning, assuming equally likely outcomes. For a fair coin, $P(\text{Heads}) = 1/2 = 0.5$ regardless of experiments.

Q3. 3. What is the difference between simulation and real-world experimentation?

Ans: Simulation uses computer-generated random numbers to model an experiment quickly and repeatedly without physical constraints. Real-world experiments involve actual physical trials, which are time-consuming but reflect true randomness and physical limitations.

Q4. 4. What are the possible outcomes of a coin toss?

Ans: A coin toss has two possible outcomes: Heads (H) or Tails (T), each with probability 0.5 for a fair coin.

Q5. 5. Why do experimental probabilities become closer to theoretical probabilities after many trials?

Ans: This is explained by the Law of Large Numbers: as the number of independent trials increases, the sample mean converges to the expected (theoretical) value. Random fluctuations average out over many trials.

Q6. 6. Which Python function is commonly used to simulate random coin tosses?

Ans: `numpy.random.choice(['H','T'], size=n)` or `numpy.random.randint(0, 2, size=n)` are commonly used. `random.choices()` from Python's standard library also works.

Q7. 7. What is a Poisson distribution?

Ans: The Poisson distribution models the probability of a given number of events occurring in a fixed interval of time or space, given the events occur independently at a constant average rate λ .

Q8. 8. In which real-life situations is Poisson distribution used?

Ans: Number of customer arrivals at a store per hour, number of emails received per day, number of traffic accidents at an intersection per week, number of server requests per minute, number of decay events in radioactive material per second.

Q9. 9. What does the parameter λ (lambda) represent in Poisson distribution?

Ans: λ (lambda) is the average rate (mean number of events) occurring in the fixed interval. It is both the mean and the variance of the Poisson distribution.

Q10. 10. What is Probability Mass Function (PMF)?

Ans: PMF gives the probability that a discrete random variable takes a specific value. For Poisson: $P(X=k) = \frac{e^{-\lambda} \times \lambda^k}{k!}$ where $k = 0, 1, 2, \dots$

Q11. 11. What is the relationship between mean and variance in a Poisson distribution?

Ans: In a Poisson distribution, the mean and variance are both equal to λ . This unique property (equidispersion) is a key characteristic of the distribution.

Q12. 12. Why do we compare empirical PMF with theoretical PMF in simulations?

Ans: Comparing empirical (simulated) PMF with theoretical PMF validates that our simulation correctly models the Poisson process. Good agreement confirms the simulation is working properly and demonstrates the Law of Large Numbers.

Practical 05 – Dice Roll & Binomial Distribution

Q1. 1. What is the theoretical probability of getting each outcome in a fair dice roll?

Ans: For a fair six-sided die, each outcome (1 through 6) has a theoretical probability of $1/6 \approx 0.1667$.

Q2. 2. What is the difference between experimental probability and theoretical probability?

Ans: Theoretical probability is derived mathematically from the assumption of equally likely outcomes. Experimental probability is calculated from actual (or simulated) trials and may differ from theoretical due to randomness, but converges to theoretical probability with more trials.

Q3. 3. How can we determine whether a dice is fair or biased?

Ans: By rolling the dice many times and performing a Chi-Square Goodness-of-Fit test comparing observed frequencies with the expected uniform distribution. If experimental frequencies significantly differ from $1/6$ each, the dice is likely biased.

Q4. 4. What is a Binomial distribution?

Ans: Binomial distribution models the number of successes in n independent trials, each with the same probability p of success. It applies when: (a) fixed number of trials n , (b) each trial is independent, (c) each trial has two outcomes (success/failure), (d) constant probability p .

Q5. 5. What do the parameters n and p represent in Binomial distribution?

Ans: n is the total number of independent trials. p is the probability of success in each individual trial.

Q6. 6. Give one real-life example of Binomial distribution.

Ans: The number of heads in 10 coin flips ($n=10$, $p=0.5$). Other examples: number of defective items in a batch, number of patients responding to a treatment, number of students passing an exam.

Q7. 7. What is the formula for the mean and variance of a Binomial distribution?

Ans: Mean (μ) = $n \times p$. Variance (σ^2) = $n \times p \times (1-p)$. Standard deviation $\sigma = \sqrt{n \times p \times (1-p)}$.

Q8. 8. Why do we compare simulated results with theoretical Binomial PMF?

Ans: Comparing simulated and theoretical PMF validates the simulation and demonstrates that the random process follows the expected distribution. It also visually confirms the Law of Large Numbers and helps understand how well the Binomial model fits the data.

Practical 06 – Sampling Techniques

Q1. 1. What is sampling and why is it important in data analysis?

Ans: Sampling is the process of selecting a subset (sample) from a larger population to make inferences about the whole. It is important because studying an entire population is often impractical due to cost, time, or size constraints. A good sample represents the population well.

Q2. 2. What is the difference between population and sample?

Ans: Population is the complete set of all individuals or items of interest. Sample is a subset selected from the population. Statistical conclusions drawn from samples are used to generalize about the population.

Q3. 3. Explain the concept of random sampling.

Ans: Random sampling is a method where every member of the population has an equal chance of being selected. This eliminates selection bias and produces a representative sample. Implemented using `numpy.random.choice()` or `pandas.DataFrame.sample()`.

Q4. 4. What is stratified sampling and when is it preferred?

Ans: Stratified sampling divides the population into subgroups (strata) based on shared characteristics and then randomly samples from each stratum proportionally. It is preferred when the population has distinct subgroups (e.g., gender, age groups) that should all be represented in the sample.

Q5. 5. What is sampling bias?

Ans: Sampling bias occurs when some members of the population are more likely to be selected than others, causing the sample to be unrepresentative of the population. It leads to incorrect conclusions and skewed statistical results.

Q6. 6. Why does stratified sampling generally give better representation of data?

Ans: By ensuring all subgroups are sampled proportionally, stratified sampling guarantees representation of each stratum. This reduces variability and produces more accurate estimates, especially when subgroups differ significantly from each other.

Q7. 7. What is the role of mean and standard deviation in data generation?

Ans: Mean (μ) defines the center of the distribution and standard deviation (σ) defines the spread. Together they parameterize distributions like the normal distribution, allowing us to generate realistic synthetic data for simulation.

Q8. 8. What is a normal distribution? Mention its important properties.

Ans: A normal (Gaussian) distribution is a continuous probability distribution that is symmetric, bell-shaped, and centered at its mean. Key properties: mean = median = mode; defined by μ and σ ; 68-95-99.7 rule (1σ , 2σ , 3σ); total area under curve = 1; extends from $-\infty$ to $+\infty$.

Q9. 9. Why is randomness important in sampling techniques?

Ans: Randomness prevents bias in sample selection, ensures each member has a fair chance of inclusion, and makes the sample statistically representative. It is the foundation of inferential statistics.

Q10. 10. Which Python libraries are commonly used for data simulation and sampling?

Ans: NumPy (numpy.random for random number generation), pandas (DataFrame.sample()), SciPy (scipy.stats for distributions), Faker (synthetic data), and scikit-learn (train_test_split for dataset splitting).

Q11. 11. What is the advantage of using pandas DataFrame in this practical?

Ans: pandas DataFrames provide easy-to-use methods for grouping (groupby), sampling (sample()), filtering, and statistical summary (describe()). They make data manipulation, stratification, and visualization straightforward.

Practical 07 – Z-Test for Hypothesis Testing

Q1. 1. What is hypothesis testing?

Ans: Hypothesis testing is a statistical method used to make decisions about a population parameter based on sample data. It involves formulating two competing hypotheses and using sample evidence to decide which is supported.

Q2. 2. What are the Null Hypothesis (H_0) and Alternative Hypothesis (H_1)?

Ans: H_0 (Null Hypothesis) is the default assumption, typically stating no effect or no difference (e.g., population mean = claimed value). H_1 (Alternative Hypothesis) is what we want to prove, stating there is an effect or difference.

Q3. 3. What is the significance level in hypothesis testing?

Ans: The significance level (α) is the threshold probability for rejecting H_0 , commonly set at 0.05 (5%). It represents the maximum acceptable probability of making a Type I error (rejecting a true H_0).

Q4. 4. Why is a Z-test used in this practical?

Ans: A Z-test is appropriate when the sample size is large ($n \geq 30$) and the population standard deviation is known. It uses the standard normal distribution to calculate the test statistic.

Q5. 5. What is meant by a two-tailed test?

Ans: A two-tailed test checks for differences in both directions (greater than or less than). $H_0: \mu = \mu_0$. The rejection region is split equally in both tails of the distribution ($\alpha/2$ in each tail).

Q6. 6. What is a p-value?

Ans: The p-value is the probability of observing a test statistic at least as extreme as the one calculated, assuming H_0 is true. A small p-value suggests the observed data is unlikely under H_0 .

Q7. 7. What is the decision rule using p-value?

Ans: If $p\text{-value} \leq \alpha$: Reject H_0 (sufficient evidence for H_1). If $p\text{-value} > \alpha$: Fail to reject H_0 (insufficient evidence against H_0).

Q8. 8. What is the formula for the Z-test statistic?

Ans: $Z = (x - \mu) / (\sigma / \sqrt{n})$, where x is the sample mean, μ is the hypothesized population mean, σ is the population standard deviation, and n is the sample size.

Q9. 9. What is the difference between sample mean and population mean?

Ans: Population mean (μ) is the true average of the entire population. Sample mean ($\bar{x}$) is the average computed from the sample. We use $\bar{x}$ to estimate μ when the full population data is unavailable.

Q10. 10. What happens if the p-value is less than the significance level?

Ans: We reject the Null Hypothesis H_0 . This means the sample provides sufficient statistical evidence that the population mean differs from the hypothesized value.

Q11. 11. Why is sample size important in hypothesis testing?

Ans: Larger sample sizes reduce the standard error ($\sigma/\sqrt{n}$), making the test more sensitive to small differences. They also ensure the sampling distribution is approximately normal, validating the use of Z or t tests.

Q12. 12. Mention some real-life applications of hypothesis testing.

Ans: Drug efficacy testing in clinical trials, quality control in manufacturing, A/B testing in websites/apps, educational research (comparing teaching methods), financial analysis (testing investment returns).

Practical 08 – T-Test

Q1. 1. What is a t-test and when is it used?

Ans: A t-test is a statistical hypothesis test used to determine if there is a significant difference between the mean of a sample and a hypothesized value (one-sample), or between two group means (two-sample). It is used when the sample size is small and/or the population standard deviation is unknown.

Q2. 2. What is the difference between a Z-test and a t-test?

Ans: Z-test is used when $n \geq 30$ and population σ is known; it uses the standard normal distribution. T-test is used when $n < 30$ or σ is unknown; it uses the t-distribution, which has heavier tails to account for additional uncertainty.

Q3. 3. Why is the t-test preferred for small sample sizes?

Ans: With small samples, the sample standard deviation s is a less reliable estimate of σ . The t-distribution accounts for this extra uncertainty by having heavier tails than the normal distribution, providing more conservative (wider) confidence intervals.

Q4. 4. What are the Null Hypothesis and Alternative Hypothesis in this practical?

Ans: H_0 : The population mean equals the hypothesized value ($\mu = \mu_0$). H_1 : The population mean is different from (or greater/less than) the hypothesized value ($\mu \neq \mu_0$ or $\mu > \mu_0$ or $\mu < \mu_0$).

Q5. 5. What is meant by a one-tailed test?

Ans: A one-tailed test checks for differences in only one direction. H_1 can be $\mu > \mu_0$ (right-tailed) or $\mu < \mu_0$ (left-tailed). The entire rejection region (α) is in one tail of the distribution.

Q6. 6. What is the formula for the t-test statistic?

Ans: $t = (\bar{x} - \mu_0) / (s / \sqrt{n})$, where $\bar{x}$ is the sample mean, μ_0 is the hypothesized mean, s is the sample standard deviation, and n is the sample size.

Q7. 7. What is the role of degrees of freedom in a t-test?

Ans: Degrees of freedom ($df = n - 1$ for one-sample t-test) determine which t-distribution to use. As df increases, the t-distribution approaches the normal distribution. Smaller df gives heavier tails, reflecting more uncertainty with smaller samples.

Q8. 8. What conclusion is drawn if the p-value is less than the significance level?

Ans: We reject H_0 . There is statistically significant evidence that the population mean differs from the hypothesized value at the chosen significance level α .

Practical 09 – Proportion Z-Test

Q1. 1. What is a proportion hypothesis test?

Ans: A proportion hypothesis test determines whether a sample proportion is significantly different from a hypothesized population proportion. It is used for categorical data with two outcomes (success/failure).

Q2. 2. When is a Z-test for proportion used?

Ans: It is used when: (a) data is categorical with two outcomes, (b) sample size n is large enough that $np \geq 5$ and $n(1-p) \geq 5$, ensuring the sampling distribution of the proportion is approximately normal.

Q3. 3. What is the difference between sample proportion and population proportion?

Ans: Population proportion (p_0) is the true proportion in the entire population (hypothesized value). Sample proportion ($\hat{p}$) is the proportion observed in the sample, calculated as number of successes / sample size.

Q4. 4. What are the Null Hypothesis and Alternative Hypothesis in this practical?

Ans: H_0 : $p = p_0$ (the population proportion equals the hypothesized value). H_1 : $p < p_0$, or $p > p_0$, or $p \neq p_0$ depending on the test direction.

Q5. 5. What is meant by a left-tailed test?

Ans: A left-tailed test checks if the parameter is significantly less than the hypothesized value. H_1 : $p < p_0$. The rejection region is in the left tail of the distribution. We reject H_0 if the test statistic is sufficiently negative.

Q6. 6. What is the formula for the Z-test statistic for proportion?

Ans: $Z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0) / n}$, where $\hat{p}$ is the sample proportion, p_0 is the hypothesized proportion, and n is the sample size.

Q7. 7. What is the meaning of p-value in hypothesis testing?

Ans: The p-value is the probability of obtaining a test statistic as extreme as or more extreme than the observed one, assuming H_0 is true. It quantifies the evidence against H_0 .

Q8. 8. What decision is taken if the calculated p-value is smaller than the significance level?

Ans: Reject H_0 . The sample provides sufficient evidence that the population proportion is different from (or less than/greater than) the hypothesized proportion.

Practical 10 – Chi-Square Test of Independence

Q1. 1. What is the Chi-Square Test of Independence?

Ans: It is a statistical test used to determine whether two categorical variables are independent of each other or whether there is a significant association between them.

Q2. 2. Why is the Chi-Square test used in this practical?

Ans: Because both variables are categorical, and we want to test if they are related. Chi-Square is the appropriate test for contingency tables with categorical data.

Q3. 3. What is a contingency table?

Ans: A contingency table (cross-tabulation) is a table showing the frequency distribution of two categorical variables. Rows represent categories of one variable and columns represent categories of the other.

Q4. 4. What are observed frequencies and expected frequencies?

Ans: Observed frequencies are the actual counts in each cell of the contingency table. Expected frequencies are the counts we would expect if the two variables were truly independent, calculated from the row and column totals.

Q5. 5. How are expected frequencies calculated in a Chi-Square test?

Ans: Expected frequency for cell (i,j) = (Row i total × Column j total) / Grand total.

Q6. 6. What are the Null Hypothesis and Alternative Hypothesis in this test?

Ans: H_0 : The two variables are independent (no association). H_1 : The two variables are not independent (there is an association).

Q7. 7. What is the formula for the Chi-Square test statistic?

Ans: $\chi^2 = \sum [(O - E)^2 / E]$, where O is the observed frequency and E is the expected frequency, summed over all cells in the table.

Q8. 8. What are degrees of freedom in a contingency table?

Ans: $df = (\text{number of rows} - 1) \times (\text{number of columns} - 1)$. For a 2×2 table, $df = 1$.

Q9. 9. What does a small p-value indicate in the Chi-Square test?

Ans: A small p-value ($< \alpha$) indicates strong evidence against H_0 , suggesting the two categorical variables are NOT independent and there is a statistically significant association between them.

Q10. 10. Mention one real-life application of the Chi-Square Test of Independence.

Ans: Testing whether gender is associated with product preference in a market survey. Other examples: association between smoking and lung cancer, relationship between education level and voting behaviour.

Practical 11 – Simple Linear Regression

Q1. 1. What is Simple Linear Regression?

Ans: Simple Linear Regression is a statistical method that models the linear relationship between one independent variable (X) and one dependent variable (Y). It fits a straight line $Y = mX + c$ that best describes the relationship.

Q2. 2. What is the purpose of regression analysis?

Ans: Regression analysis is used to understand and quantify the relationship between variables and to make predictions. It helps determine how changes in independent variables affect the dependent variable.

Q3. 3. What are dependent and independent variables in this practical?

Ans: The independent variable (X, predictor) is the input feature, such as study hours. The dependent variable (Y, response) is the output we want to predict, such as exam scores.

Q4. 4. What does the slope of the regression line represent?

Ans: The slope (m) represents the change in the dependent variable Y for a unit increase in the independent variable X. For example, if $m = 5$, then for every 1-unit increase in X, Y increases by 5 units.

Q5. 5. What does the intercept represent in a regression equation?

Ans: The intercept (c or b_0) is the predicted value of Y when $X = 0$. It is the point where the regression line crosses the Y-axis.

Q6. 6. What is the general form of a simple linear regression equation?

Ans: $Y = mX + c$ (or $Y = \beta_1 X + \beta_0$), where m (or β_1) is the slope and c (or β_0) is the intercept.

Q7. 7. What is the meaning of positive correlation?

Ans: Positive correlation means as X increases, Y also tends to increase. The slope of the regression line is positive. Correlation coefficient r is between 0 and +1.

Q8. 8. Why is a scatter plot used before fitting a regression model?

Ans: A scatter plot visually reveals whether a linear relationship exists between X and Y, identifies outliers, and shows the direction and strength of the relationship, helping validate that linear regression is an appropriate model.

Q9. 9. What is RMSE and why is it calculated?

Ans: RMSE (Root Mean Square Error) = $\sqrt{(\sum(Y_{\text{actual}} - Y_{\text{predicted}})^2 / n)}$. It measures the average magnitude of prediction errors in the same units as Y. Smaller RMSE indicates a better-fitting model.

Q10. 10. Which Python libraries are commonly used for regression analysis?

Ans: scikit-learn (LinearRegression), NumPy (np.polyfit), pandas (data handling), Matplotlib/Seaborn (visualization), SciPy (scipy.stats.linregress), and Statsmodels (detailed statistical output).

Q11. 11. How can regression be used in real-life business or prediction problems?

Ans: Predicting house prices from size/location, forecasting sales from advertising spend, estimating crop yield from rainfall, predicting patient outcomes from treatment dosage, financial market price prediction.

Practical 12 – Directional Derivatives

Q1. 1. What is a directional derivative?

Ans: A directional derivative measures the rate of change of a function at a given point in a specified direction. It generalizes the concept of a partial derivative to any direction in space.

Q2. 2. What is meant by the gradient of a function?

Ans: The gradient (∇f) of a scalar function is a vector containing all its partial derivatives. For $f(x,y)$: $\nabla f = (\partial f / \partial x, \partial f / \partial y)$. It points in the direction of the steepest ascent.

Q3. 3. Why is the gradient vector important in directional derivatives?

Ans: The directional derivative in direction $\hat{u}$ is computed as the dot product of the gradient and the unit vector: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. The gradient thus encodes all directional information needed.

Q4. 4. What is a unit vector and why is it used?

Ans: A unit vector has magnitude 1. It is used to represent a pure direction without any scale. In directional derivatives, we normalize the direction vector so that the rate of change is measured per unit distance.

Q5. 5. What does the directional derivative represent physically?

Ans: It represents how fast the function value changes as we move from a given point in a specific direction. For example, in terrain mapping, it represents the slope in a chosen direction.

Q6. 6. What is the formula for directional derivative?

Ans: $D_{\hat{u}} f(x,y) = \nabla f \cdot \hat{u} = (\partial f / \partial x)u_x + (\partial f / \partial y)u_y$, where $\hat{u} = (u_x, u_y)$ is the unit direction vector.

Q7. 7. What is the difference between partial derivative and directional derivative?

Ans: A partial derivative measures the rate of change along a coordinate axis direction (x-axis or y-axis only). A directional derivative measures the rate of change in any arbitrary direction.

Q8. 8. In which direction is the directional derivative maximum?

Ans: The directional derivative is maximum in the direction of the gradient ∇f . The maximum value is $\|\nabla f\|$ (the magnitude of the gradient).

Q9. 9. Which Python library is commonly used for symbolic differentiation?

Ans: SymPy is used for symbolic mathematics in Python. It can compute partial derivatives, gradients, and directional derivatives analytically.

Q10. 10. What is the purpose of normalizing a vector before finding directional derivative?

Ans: Normalization ensures the direction vector is a unit vector (magnitude = 1), so the directional derivative gives the true rate of change per unit distance. Without normalization, the result would be scaled by the vector's magnitude.

Q11. 11. Mention one real-life application of directional derivatives.

Ans: In machine learning, gradient descent uses gradients (directional derivatives) to find the minimum of a loss function. In physics, directional derivatives describe electric field intensity and heat flow direction.

Practical 13 – Gradient Descent

Q1. 1. What is Gradient Descent?

Ans: Gradient Descent is an iterative optimization algorithm used to minimize a function (typically a loss/cost function) by repeatedly moving in the direction of the steepest descent (negative gradient).

Q2. 2. Why is Gradient Descent used in machine learning?

Ans: Most ML models are trained by minimizing a loss function (e.g., MSE for regression, cross-entropy for classification). Gradient Descent efficiently finds the parameter values that minimize this loss, making model training feasible even for large datasets.

Q3. 3. What is a loss function?

Ans: A loss function (cost function) measures how wrong the model's predictions are compared to actual values. Common examples: Mean Squared Error (MSE) for regression, Cross-Entropy Loss for classification.

Q4. 4. What is meant by minimizing a loss function?

Ans: Minimizing a loss function means finding the model parameters (weights, biases) that make the predictions as close as possible to the true values, resulting in the smallest possible error.

Q5. 5. What is the role of the learning rate in Gradient Descent?

Ans: The learning rate (α) controls the step size at each iteration. It determines how much we update the parameters in the direction of the negative gradient. Choosing the right learning rate is critical for convergence.

Q6. 6. What happens if the learning rate is too large?

Ans: If α is too large, the algorithm may overshoot the minimum, causing the loss to oscillate or diverge. The algorithm fails to converge.

Q7. 7. What happens if the learning rate is too small?

Ans: If α is too small, convergence is very slow, requiring many iterations. The algorithm may get stuck in local minima or take an impractically long time to train.

Q8. 8. What is the gradient of a function?

Ans: The gradient is a vector of partial derivatives of the loss function with respect to each parameter. It points in the direction of the steepest increase, so we move in the opposite direction (negative gradient) to minimize the loss.

Q9. 9. What is the update rule used in Gradient Descent?

Ans: $\theta = \theta - \alpha \times \nabla J(\theta)$, where θ represents the parameters, α is the learning rate, and $\nabla J(\theta)$ is the gradient of the loss function J with respect to θ .

Q10. 10. What is the purpose of iterations in Gradient Descent?

Ans: Each iteration updates the parameters by one step in the direction of the negative gradient. Multiple iterations are needed because a single step rarely reaches the minimum; we converge gradually over many steps.

Q11. 11. How do we know that Gradient Descent has converged?

Ans: Convergence is indicated when the change in loss between iterations falls below a small threshold (tolerance), the gradient magnitude becomes nearly zero, or after a fixed maximum number of iterations. A loss-vs-iteration plot flattening out visually shows convergence.

Q12. 12. Mention one real-life application of Gradient Descent in AI or data science.

Ans: Training neural networks (backpropagation uses gradient descent to update weights), linear/logistic regression optimization, recommendation systems, image recognition model training, and natural language processing model fine-tuning.